

01 / 06 Overview

Rayforce Cloud and market data

An overview of the engine, the proposed cloud service and where external data fits into the architecture.

The proposal is to provide managed analytical workspaces with optional market data services. Clients can query connected datasets alongside their own data.

01 Rayforce

The database and analytics engine.

Existing technology

02 Rayforce Cloud

A managed environment for storage, queries and analytical workloads.

Proposed service

03 Market data services

Optional datasets, maintained and made available for queries within a workspace.

Proposed integration

What is Rayforce?

An embeddable analytics engine written in C, combining columnar processing and graph operations.

Tabular analytics

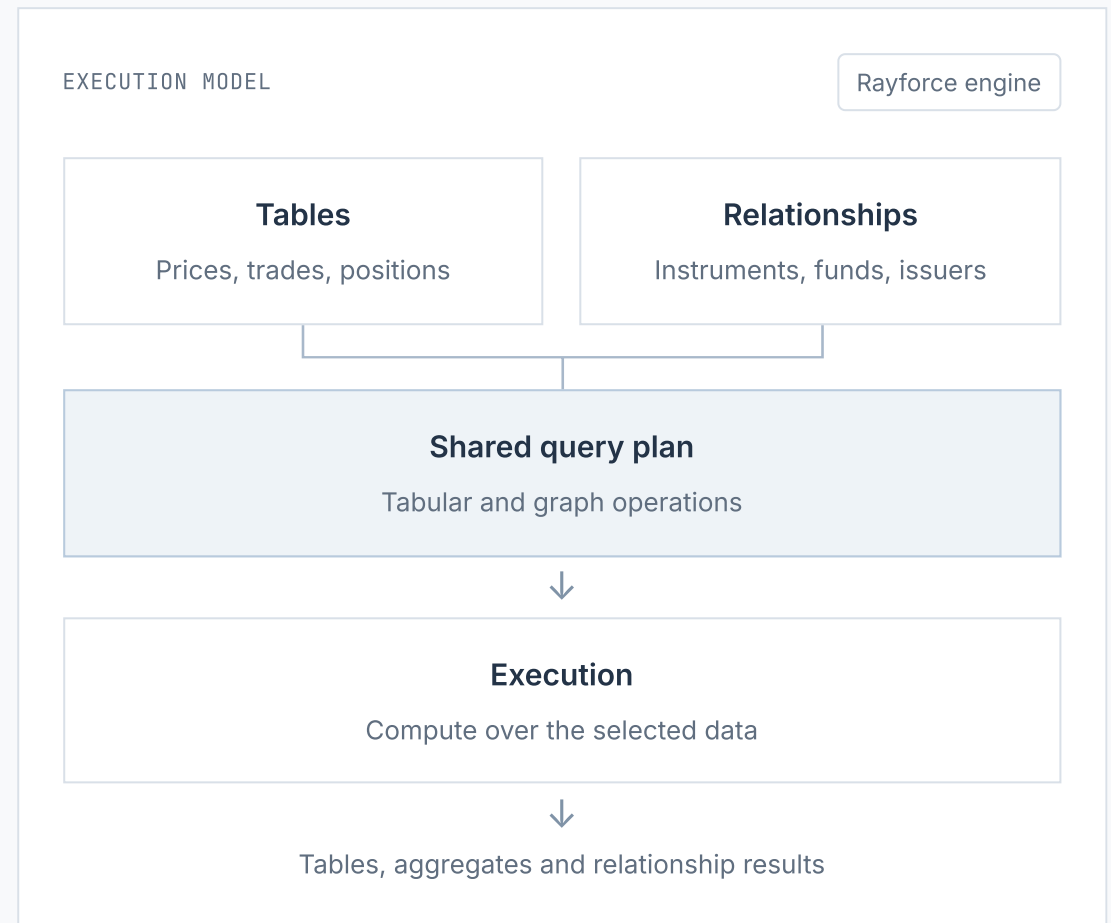
Filters, joins, aggregations and window functions over structured data.

Relationship analysis

Graph traversal and algorithms for connected entities, such as instruments, funds and issuers.

Programmatic access

The Rayfall language for queries and a C API for embedding the engine in applications.



A workspace with optional market data

Rayforce Cloud would manage compute and storage. Clients would choose the datasets and analyses they need.

01 Create a workspace

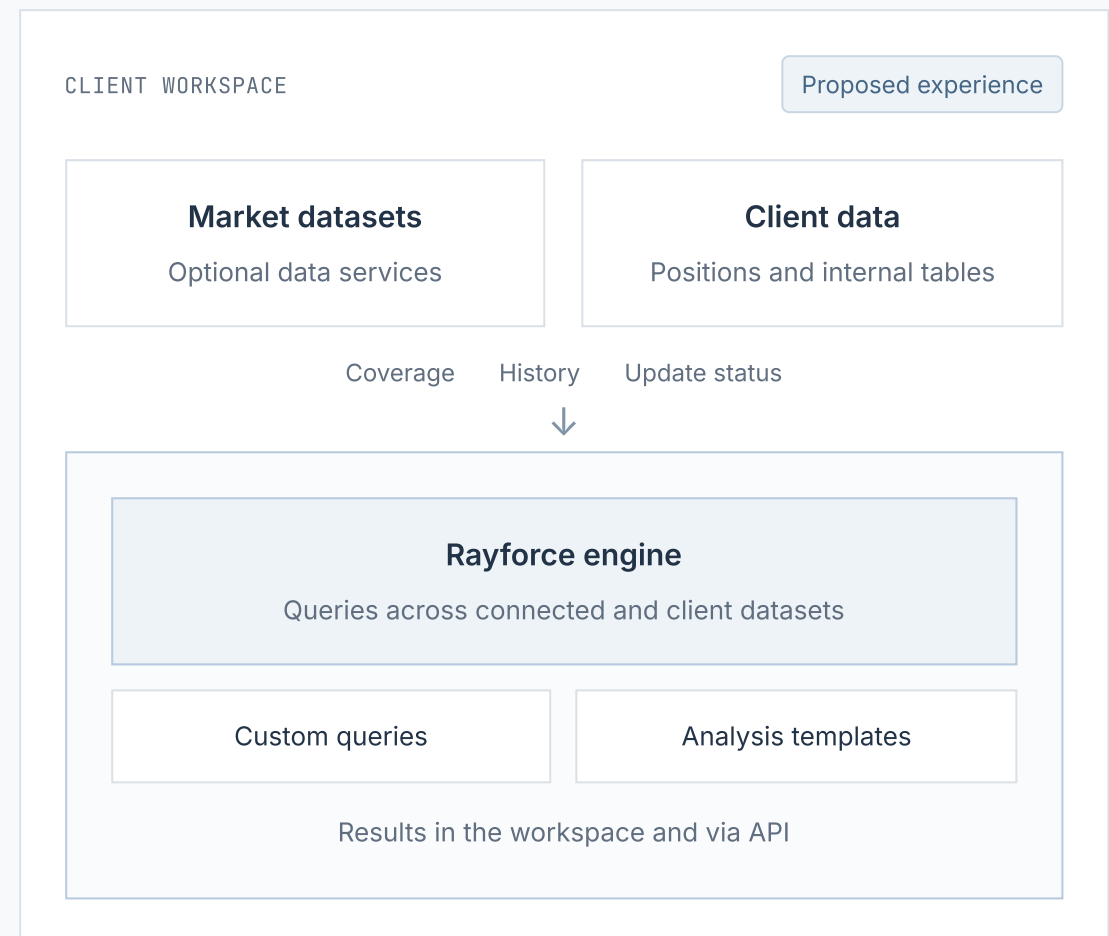
A managed environment for the client's data, queries and analytical workloads.

02 Connect datasets

Review coverage, history and update frequency. Connect entitled market datasets and add internal data.

03 Run an analysis

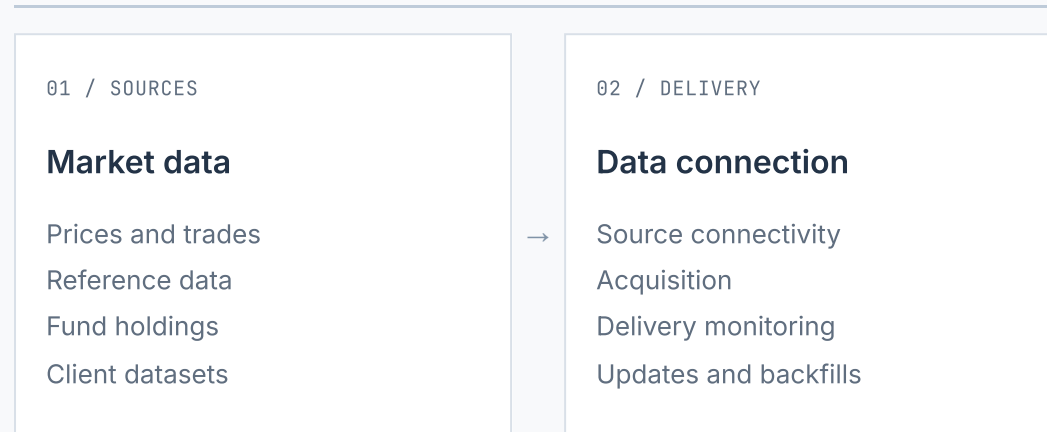
Write custom queries or use maintained analysis templates. Access results through the workspace or API.



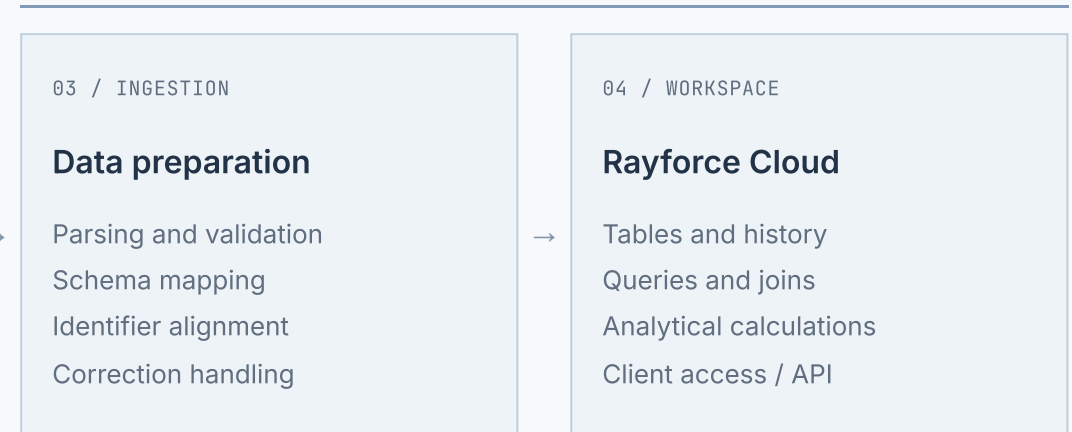
Where market data connects

A delivery service brings data from external sources. An ingestion layer makes it available as datasets in Rayforce Cloud.

External sources and delivery



Integration into Rayforce Cloud



INTEGRATION BOUNDARY A delivery connector feeds the ingestion layer, which publishes prepared datasets to a client workspace.

Source-specific delivery

Files, APIs or streams can be considered according to the source. Each requires an appropriate connector and update policy.

Provider-independent design

The delivery layer could be supplied by a data partner or connected directly. Data access follows the client's entitlements.

Working with connected data

Once datasets are available in a workspace, clients can combine market data with their own queries and positions.

ANALYSIS	INPUTS	COMPUTATION
Market activity	Prices, trades and instrument reference data	Returns, volumes, price ranges and time-window comparisons
Portfolio valuation	Client positions, prices and exchange rates	Position values and aggregation by instrument, currency or issuer
Portfolio look-through	Positions, fund holdings and instrument-to-issuer relationships	Direct and indirect exposure through nested holdings

Custom queries or reusable analyses

The service could include maintained query templates alongside the client’s own calculations.

Data context with each result

Source dates, coverage and update status should remain visible when interpreting an analysis.

The proposed service model

Rayforce Cloud would provide managed analytical workspaces with optional market data services. Clients could use their own data, connected datasets, or both.

RAYFORCE

The independent engine

The database and computation layer, also available for use outside the cloud service.

RAYFORCE CLOUD

Managed workspaces

Compute, storage, query execution and client access, with support for custom and reusable analyses.

MARKET DATA SERVICES

Optional connected datasets

Data supplied and maintained through integrations with providers or delivery partners.

Integration details

Source coverage · delivery method · schema and identifiers · update frequency · historical data and corrections · hosting and support

Cloud and integration capabilities described here are a proposed service design.

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